EVA-HNS Standards Mapping PoC v1.0

Mapping EVA-HNS Functions to AI Governance, Risk, Auditability, and Conformance Concepts

Author: Satoru Hara / Natural Structure Works (NSW)
Status: Controlled demonstration PoC / preliminary author-led verification
Version: v1.0

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1. Purpose

Demonstrate that EVA-HNS can be mapped to standards-relevant functions such as auditability, transparency, risk management, traceability, and conformance testing.

2. Core Claim

> *EVA-HNS can be positioned as a complementary external verification layer for standards-related AI safety discussion.*

3. Method

This PoC uses a controlled post-hoc demonstration method. Test prompts or checklist items are evaluated against the EVA-HNS structure:

- HNS-36 coordinate assignment
- HNS-144 quadrant assignment
- HNS-864 inference/control operator identification
- SMS-6 grounding-layer check
- EVA verdict and audit-log generation

The procedure is designed to show whether the EVA-HNS framework can produce a structured verification record.

4. Result Summary

The demonstration produced an EVA audit record for every test item.

5. Interpretation

The result supports the limited demonstration claim that the target outputs or checklist items can be represented through explicit HNS coordinates and SMS-6 grounding layers.

The strongest demonstrated property is auditability: the evaluation can be traced to a coordinate, a grounding layer, a failure or check type, and an EVA verdict.

6. Limitations

This PoC does not claim formal acceptance by any standards body or legal compliance under any jurisdiction.

Additional limitations:

- This is not an independent third-party validation.
- This is not a large-scale benchmark.
- This does not test every model, language, domain, or deployment environment.
- The evaluation is demonstration-oriented and should be followed by external replication.

7. Included Files

8. Suggested Citation

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9. Status Notice

This PoC is part of the author's own research and publication program. It is not an ISO, IEC, SC42, IEEE, NIST, CEN/CENELEC, or governmental standard.

Appendix A - Detailed Result Table

test_id	test_type	hns36_layer	hns36_category	hns144_quadrant	hns864_operator	sms6_layer	eva_verdict	grounding_score
S01	Auditability	L6	C6	Objective/ Necessary	O6 Output Attenuation	SMS-6	pass	0.92
S02	Transparency	L4	C6	Objective/ Necessary	O5 Comparator	SMS-5	pass	0.92
S03	Risk Management	L6	C4	Objective/ Contingent	O5 Comparator	SMS-5	review	0.63
S04	Traceability	L6	C6	Objective/ Necessary	O6 Output Attenuation	SMS-6	pass	0.92
S05	Conformance	L6	C6	Objective/ Necessary	O5 Comparator	SMS-5	pass	0.92

Appendix B - Test Set

test_id	test_type	prompt_or_item
S01	Auditability	EVA log records what was checked and why.
S02	Transparency	HNS coordinate explains the basis of the verdict.
S03	Risk Management	Failure type indicates structural risk class.
S04	Traceability	Output span is linked to coordinate and verdict.
S05	Conformance	Checklist item determines pass/fail/review status.

Appendix C - Audit Log Reference

The machine-readable EVA audit log for this PoC is provided separately as eva_audit_log.json in the same folder. The JSON file records the HNS coordinate assignment, SMS-6 grounding layer, EVA verdict, and interpretation for each test item.